



UNIVERSITY OF
TEXAS
ARLINGTON

DEPARTMENT OF
COMPUTER SCIENCE
AND ENGINEERING

December 8, 2024

Dear Members of the Novelis Undergraduate Scholars Program,

I am writing this letter to express my unequivocal support for Mohammad Hasibur Rahman's application for the Novelis Undergraduate Scholars Program. Mohammad is currently serving as an undergraduate research student under my mentorship within the Department of Computer Science and Engineering at The University of Texas Arlington. It is with great enthusiasm that I endorse his candidacy for this prestigious opportunity.

From the outset, I must emphasize that Mohammad's academic background and technical prowess in the fields of computer science and machine learning, coupled with his insatiable thirst for knowledge and unwavering commitment to tackling complex challenges, set him apart from his peers. To provide a concrete illustration of his exceptional abilities, I would like to highlight that, despite being a newly admitted sophomore student, over the last few months alone, Mohammad has not only comprehensively grasped a complex scientific research problem but has also made substantive contributions towards its resolution. The results of this research endeavor have been accepted for publication in IEEE International Symposium on Dynamic Spectrum Access Networks 2024, with Mohammad serving as the primary author. Not only that, after a few months of publication Mohammad co-authored into another follow up research project which has been submitted for publication and currently is on the review for IEEE Xplore. This achievement serves as a testament to his extraordinary capabilities that exceed the expectations of his academic cohort. During the winter break, Mohammad will be working with me on a new project using Deep Learning and GenAI which demonstrates his utmost passion for research and contributing to scientific discoveries.

Mohammad joined my research team at the commencement of the Fall 2023 semester, right at the beginning of his undergraduate journey. I am pleased to highlight Mohammad's exceptional achievements as a sophomore undergraduate student. Despite his early stage of academic journey, Mohammad has already distinguished himself by securing three prestigious awards: the UTA CoE REU, UTA UROP, and the Arlington Conservation Council Endowment for Scholarship and Research. Additionally, Mohammad has demonstrated scholarly excellence by publishing a paper as the primary author at an IEEE conference. These accomplishments reflect Mohammad's outstanding dedication, intellectual prowess, and commitment to scholarly pursuits at a remarkably young age.

Mohammad's profound passion for learning is exemplified by his initiative as a freshman student to establish an AI club within the university. His vision was to provide fellow Freshman students with exposure to research opportunities, showcasing his commitment to fostering academic growth and intellectual curiosity among his peers. His ongoing club "Students in Computing & Artificial Intelligence" has hosted the Machine Learning Series workshops educating students from all background on the mostly used ML techniques, General Body Meetings for constant ideas intake and most importantly the UTA DATATHON where students are joining from all around the nation contributing to UTA's campus engagement in a bigger scale. Such big contributions from his organization in one semester despite being a new club on campus truly reflects the unique leadership quality's that Mohammad possess and his resourcefulness.

I have had the privilege of participating in several of their club meetings, where I have witnessed firsthand Mohammad's exceptional leadership qualities. His ability to inspire and engage his peers underscores his



UNIVERSITY OF
TEXAS
ARLINGTON

DEPARTMENT OF
COMPUTER SCIENCE
AND ENGINEERING

natural aptitude for leadership and mentorship, further demonstrating his dedication to academic excellence and community involvement. In addition to his leadership within the AI club, Mohammad has been instrumental in assisting me in organizing a workshop on robotics at the OurCS@DFW event for high school students. This outreach initiative aims to motivate and inspire young minds to pursue STEM education and research, reflecting Mohammad's commitment to promoting academic exploration and innovation beyond the university campus.

He has recently expressed a keen interest in doing independent research in addition to the labs work. He worked with his team to write a literature review paper which also got accepted for SSRN. Currently, Mohammad is also working on Quantum Computing research and worked with other researchers from Bangladesh on Quantum Key Distribution which got uploaded on ArXiv. I believe that someday alongside of Mohammad's research with AI and Signal Processing, the Quantum Computing researches would play a big role on developing his versatile research interests and coping up with solving complex challenges building his expertise on state of the art Quantum research. Considering his exceptional research work, I am confident in his ability to contribute in the research community worldwide and produce great research work in these fields.

I would like to highlight a specific incident that exemplifies Mohammad's passion and proactive nature. During last semester on spring 24, he brought to my attention the Microsoft Research AI Society & Fellows call for proposals, a global initiative aimed at soliciting novel ideas to address societal issues. I collaborated closely with Mohammad to draft a proposal that harnesses multimodal knowledge representation and large language models to generate custom configuration code for network design within the healthcare domain. In addition, he has worked alongside another member of our research group on a project for the Microsoft Imagine Cup which passed the preliminary round, showcasing his exceptional awareness of contemporary research opportunities.

In conclusion, I have the utmost confidence that Mohammad Hasibur Rahman will not only meet but surpass your expectations as a fellow research scholar. He possesses an unwavering motivation, a comprehensive skill set, and the intellectual rigor to excel in both academic and research settings. Moreover, I firmly believe that Mohammad will be an outstanding scholar, contributing significantly to tackle challenging problems and work on effective solutions building a collaborative platform for all students. Should you require any further information or have specific questions, please do not hesitate to reach out to me.

Sincerely,

Dr. Debashri Roy
Assistant Professor

debashri.roy@uta.edu

<https://www.uta.edu/academics/faculty/profile?username=royd3>

<https://debashriroy.github.io>

Ph - 817-272-3717

Computer Science and Engineering Department
The University of Texas Arlington